

Physics 1
Constants

Constants

Wien's displacement constant	$a = 2.90 \times 10^{-3} \text{ K m}$
Boltzmann's constant	$k = 1.38 \times 10^{-23} \text{ J/K}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Planck's constant	$h = 6.63 \times 10^{-34} \text{ J s}$
Bohr's constant	$B = 2.18 \times 10^{-18} \text{ J}$
Speed of light in vacuum	$c = 3.00 \times 10^8 \text{ m/s}$
Elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
Acceleration of gravity	$g = 9.81 \text{ m/s}^2$
Atomic mass unit	$u = 1.66 \times 10^{-27} \text{ kg}$
Light year	$1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$

Masses

Electron	$5.4858 \times 10^{-4} \text{ u} = 9.1094 \times 10^{-31} \text{ kg}$
Proton	$1.007\,276 \text{ u} = 1.6726 \times 10^{-27} \text{ kg}$
Neutron	$1.008\,665 \text{ u} = 1.6749 \times 10^{-27} \text{ kg}$
Hydrogen	$1.007\,825 \text{ u}$
Deuterium	$2.014\,102 \text{ u}$
Tritium	$3.016\,049 \text{ u}$
Helium-3	$3.016\,029 \text{ u}$
Helium-4	$4.002\,603 \text{ u}$
Alpha particle	$4.001\,506 \text{ u}$

Physics 1
Formulas

Motion and forces

$v = at + v_0$	$s = \frac{v + v_0}{2} t$	
$v^2 - v_0^2 = 2as$	$s = \frac{1}{2}at^2 + v_0t$	
$\Sigma F = 0 \iff v \text{ constant and rectilinear}$		
$\Sigma \vec{F} = m \vec{a}$	$F^* = F$	$G = mg$
$R = \mu N$	$B = \rho_f Vg$	$L = kv^2$
$\vec{p} = m \vec{v}$	$\Sigma \vec{F} \cdot t = \Delta \vec{p}$	
$m_A v_A + m_B v_B = m_A v_{A0} + m_B v_{B0}$		

Mechanical energy and power

$W = Fs \cos \theta$	$W_{\Sigma F} = \Delta E_k$	$P = W/t$
$E_k = \frac{1}{2}mv^2$	$E_p = mgh$	$E = E_k + E_p$
$\frac{1}{2}mv^2 + mgh = \frac{1}{2}mv_0^2 + mgh_0$		
$\eta = \frac{\text{useful energy}}{\text{supplied energy}} = \frac{\text{useful power}}{\text{supplied power}}$		

Pressure and temperature

$p = F/A$	$T = t + 273$	$E_k = \frac{3}{2}kT$
$c = \frac{Q}{m \Delta T}$	$C = \frac{Q}{\Delta T}$	$\Delta U = W + Q$

Earth

Mass	$5.972 \times 10^{24} \text{ kg}$
Mean radius	6371 km
Mean distance from the Sun	$1.496 \times 10^{11} \text{ m}$
Solar irradiance outside the atmosphere	1361 W/m^2

Sun

Mass	$1.99 \times 10^{30} \text{ kg}$
Radius	$6.96 \times 10^8 \text{ m}$
Surface temperature	5780 K
Radiated power	$3.83 \times 10^{26} \text{ W}$

SI prefixes

10^{-18}	atto	a	10^{-3}	milli	m	10^3	kilo	k
10^{-15}	femto	f	10^{-2}	centi	c	10^6	mega	M
10^{-12}	pico	p	10^{-1}	deci	d	10^9	giga	G
10^{-9}	nano	n	10^1	deca	da	10^{12}	tera	T
10^{-6}	micro	μ	10^2	hecto	h	10^{15}	peta	P

Pressure and temperature (cont.)

$$COP = \frac{Q_{\text{out}}}{E_{\text{electric}}} \qquad COP_{\text{Carnot}} = \frac{T_{\text{out}}}{T_{\text{out}} - T_{\text{in}}}$$

Waves and radiation

$f = 1/T$	$v = \lambda f$	$U = P/A$
$\lambda_{\text{peak}} \cdot T = a$	$U = \sigma T^4$	$I = \frac{P}{4\pi r^2}$

Atomic and nuclear physics

$E = hf = \frac{hc}{\lambda}$	$E_n = -\frac{B}{n^2}$	
$hf = E_n - E_m, \quad n > m$	$hf = B \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$	
$A = Z + N$	$E_0 = mc^2$	$\Delta E_0 = \Delta m \cdot c^2$

Electricity

$I = \frac{Q}{t}$	$U = \frac{W}{Q}$	$R = \frac{U}{I}$
$W = UIt$	$P = UI = RI^2 = \frac{U^2}{R}$	
Series:	$R_{\text{tot}} = R_1 + R_2 + \dots + R_n$	
Parallel:	$R_{\text{tot}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} \right)^{-1}$	